

Problem#01: Bool Function and String

Define a function `bool checkPalindrome(string str)` that takes a string variable as parameter, and returns **true** if string is palindrome otherwise return false..

Problem#02: Void Function and 2D arrays

Write a C++ program that creates a function **subtractMatrices** which receives **three parameters** as 2D arrays of float types named first, second, and third. Function will add the values of first and second matrix and store the result in third matrix. Also write a function **print(float arr2D[][])** and call it in the main program to print the values of third array. Note: Consider all 2D arrays have equal size of rows and cols i.e. **5x5** and must initialize/input 2D arrays in the main program.

Problem#3: Void Function and Pass-by-Reference

Write a function **precedingChars** that takes **3** parameters as alphabet characters and find their preceding alphabets. For example if user passed the characters: **a, m, y** then it will return **b, n, z**. Take the input in main, pass values to function and must show the output in main. After the function call, actual parameters should be updated in the main program.

Problem#04: Void and Pass-by-Reference

Write a function **"lower_to_Upper"**. It will take 4 alphabets as parameters in lowercase and will convert them into uppercase. Don't use any library function. Display the changed values in main.

Problem#05: Void - Function , 2D array and Constant parameters

Write a function **"reverse2D"** which will take two parameters 1) 2D array of char type and 2) columnSize. This function will only print the values of 2D array in reverse order and returns nothing. Also make sure that user cannot change the values of 2D array inside the reverse2D function.

Problem#06: Value-Returning Function

Write the definition of a void function named **"gradePFCourse"** that takes as input **three** integer numbers. The function computes the sum and average of the three numbers. If the average is greater than or equal to 70, it returns **"Pass"**; otherwise it returns **"Fail"**.

Problem#07: Sum of the Series

Write a C++ program to find the sum of the following series by use of user-defined functions.

$$\text{Sum} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$

1. Write a function named **"power"** having two parameters **"X"** and **"n"** of **int** types
2. Write a function named **"factorial"** with one **int** parameter **"X"**
3. Write a function named **"sumOfSeries"** having two parameters of int type **"X"**, **"n"**. Use the above **power** and **factorials** functions inside **sumOfSeries** to calculate sum of series up to n terms.

Problem#08: Value-Returning-Function

Write a function `int maxSumOfDigits(int num1, int num2)` that takes two integers and calculates the sum of all digits in num1 and num2. After find both sums, it then returns the maximum sum. Also write function `int max(int sum1, int sum2)` and call it inside the `maxSumofDigits` function to find max.